

9 Special Effects

3ds Max is used as a special effects tool in the film, video, and game industries. Special effects can be created in various ways, but many of the best special effects simulate reality. 3ds Max has tools for creating and animating objects that respond to real-world forces such as gravity. Objects animated in this way can appear highly realistic and can seamlessly blend with real-world scenes.

Particles simulate the motion of large numbers of objects and are terrific for all sorts of fire, water, and atmospheric effects. 3ds Max has several basic types of particle systems, as well as Particle Flow, which offers much more sophisticated tools. Reactor's dynamics allow objects to animate and collide with each other by using forces, and soft body dynamics allow physical forces to affect the actual shape of an object.

9.1 Particle Systems

Particle systems are used to simulate all sorts of natural phenomena, from smoke and fire to rain, sparks, and any other effect that requires animating a large number of objects. In 3ds Max, there are several types of particle systems. Basic particle systems allow for simple simulations such as a spray or cloud of particles. These basic systems can be affected by forces so they can interact with an environment. For more-sophisticated applications, Particle Flow is an event-driven system that allows for a high degree of control over particle behaviour.

9.1.1 Basic Particle Systems

3ds Max has several types of basic particle systems, also called non-event particle systems.

These provide straightforward tools for creating simple effects such as snow, rain, exhaust, water sprays, and more. These particle systems are located in the Create tab.

The Spray particle system emits particles from a simple plane in a spray like pattern. Super Spray is a much more robust version of Spray and allows for control of spray direction. Snow simulates falling snow or confetti like particles that fall from a plane in a random pattern. Blizzard is a more robust version of snow and can create more types of particles, including meta particles and instanced geometry. Particle Array uses geometry as the